Genome Assembly

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Reconstructing a genome from fragments

Mapping compares an isolate to a reference.

But many public-health tasks need the isolate's own complete genoma, independent of any reference. Genome assembly stitches the reads back together into long, contiguous sequences, capturing the full genome including the accessory parts that mapping misses.

What this module covers

- Why assembly is needed
- Short, long vs. hybrid data
- Two assembly algorithms
- Why repeats are a problem
- From reads to an annotated genome
- Quality metrics
- Applications

The big idea: reassemble a shredded document

Imagine many copies of a book, shredded into millions of scraps. You rebuild the text by spotting where scraps overlap.

The twists?

Repeats & read errors make it hard.

THE RAW MATERIAL

Short vs long reads for assembly

Short reads (Illumina)

- 100–300 bp, very accurate, cheap
- Cannot span repeats longer than a read
- Bacterial assemblies stay in many pieces
- Never fully closed on their own

Long reads (PacBio / ONT)

- 10–30 kbp+ — span most bacterial repeats
- Can close a chromosome into one contig
- HiFi / R10 now reach Q20+ accuracy
- Portable; field-deployable (MinION)

Read length is what resolves repeats — the key to a contiguous assembly.

THE RAW MATERIAL

Hybrid strategies combine the best of both

Long reads provide the scaffold and resolve repeats; short reads provide per-base accuracy.
Combining them = most reliable route to a finished genome.

Long-first, then polish

- Assemble long reads → backbone
- Polish with accurate short reads
- Tools: Pilon, Polypolish

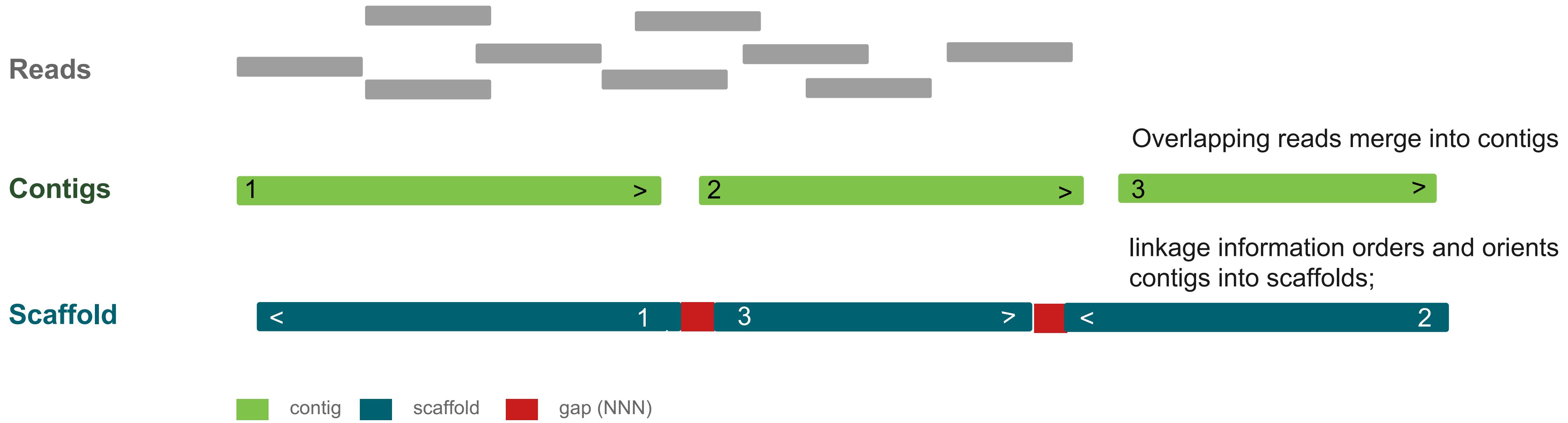
Short-first, then scaffold

- Assemble short reads → contigs
- Order/join them with long reads
- Tools: Unicycler, SPAdes hybrid

VOCABULARY

Contigs → scaffolds → gaps

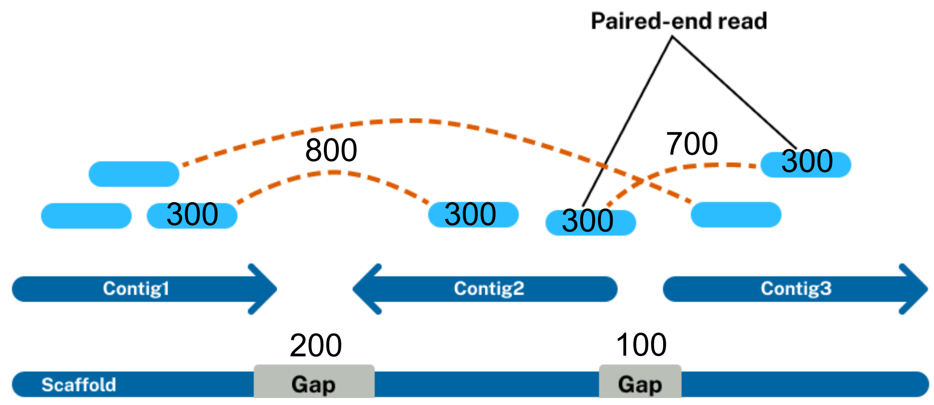
; unknown stretches are filled with N.

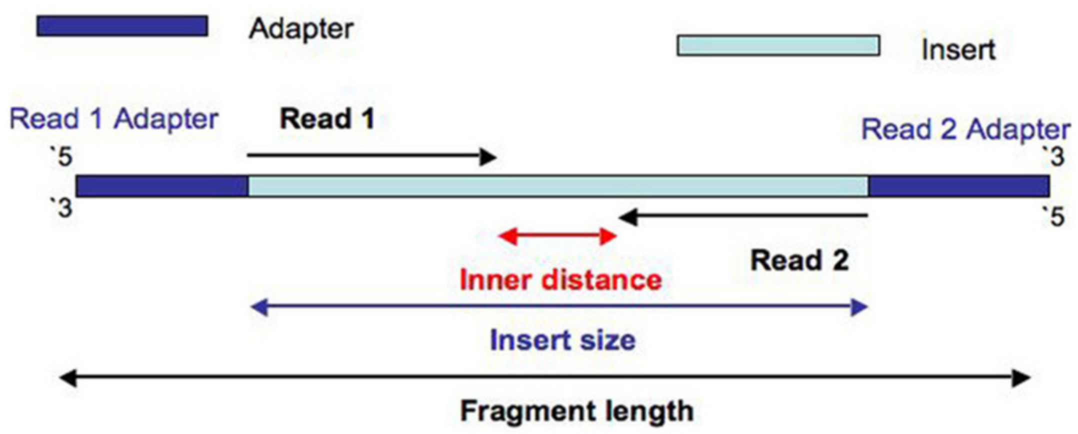


A fully “closed” genome is a single contig per replicon — chromosome plus any plasmids.

Single-end reads

Paired-end reads





Paired-end sequencing provides:

Ordering and orientation: If a read aligns forward on one contig and its mate aligns in reverse, assemblers can determine in which direction they face.

Known distance: You sequence both the forward and reverse ends of a single, larger DNA fragment. While the middle of the fragment goes unsequenced, you know the approximate length of the insert.

ALGORITHMS

Assembly = build a graph, then find a path

Every assembler **turns the heap of reads into a graph**, then **finds a sensible path through it**.

Two paradigms have shaped the field. They differ in which read type they suit, and how the graph is built.

Overlap–Layout–Consensus

for long reads

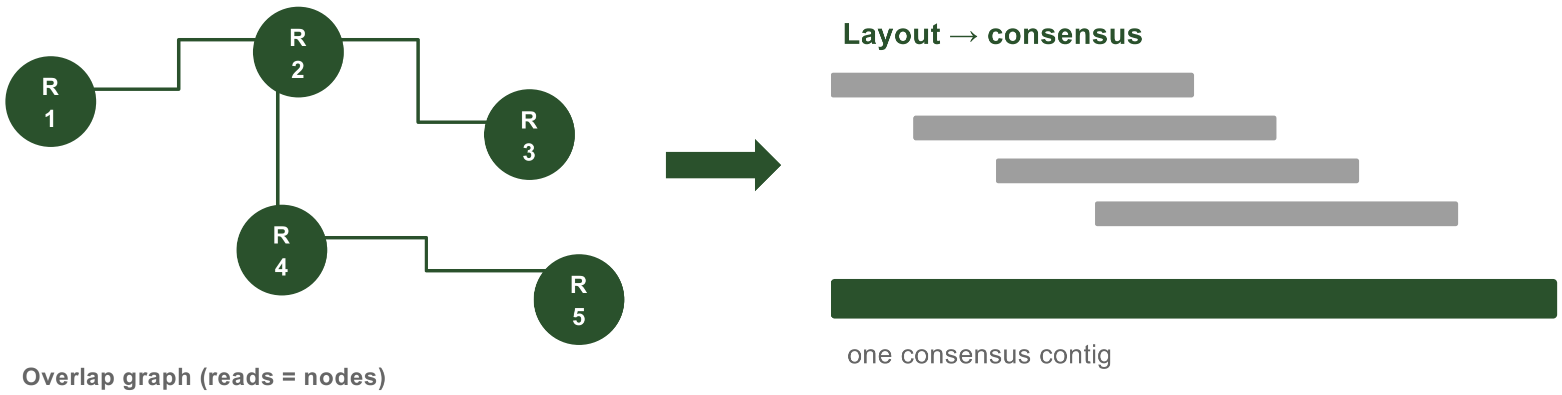
De Bruijn graph

for short reads

ALGORITHMS

Overlap–Layout–Consensus (OLC)

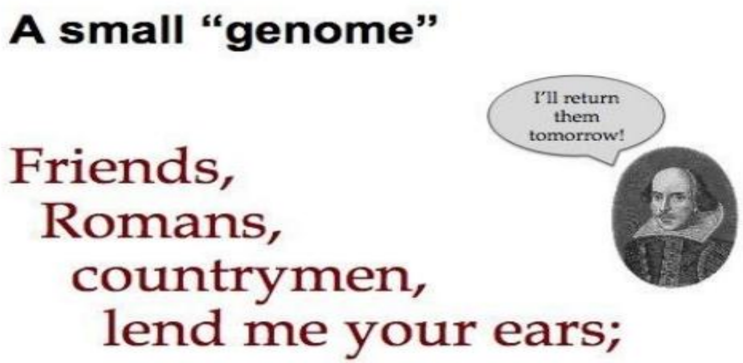
Compare reads to each other, lay them out, agree a consensus.

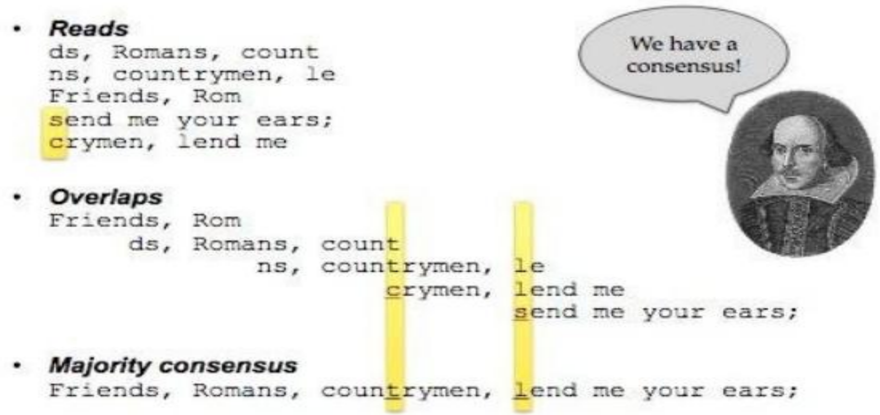


Scales with the square of read count → suited to fewer, longer reads

Tools: Canu, Flye, hifiasm, miniasm.

OLC and Hamiltonian paths





The OLC consensus step: overlapping reads are aligned to extract a majority-vote sequence

Key insight: OLC finds a Hamiltonian path (each full read is visited— computationally hard).

ALGORITHMS

De Bruijn graph & k-mers

For thousands of short reads, all-vs-all overlap is impossible. Instead, break reads into fixed-length k-mers and link them.

One read A G C T G A A C

→ overlapping k-mers (k-1)

AGC | GCT | CTG | TGA | GAA | AAC

→ nodes linked into a path

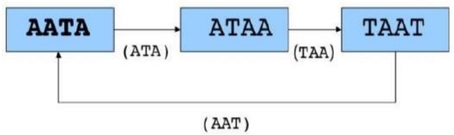
Optimal k size = structural clarity and data sensitivity
Tools: Unicycler, SPAdes, SKESA, Shovil

A de Bruijn graph decomposes each read into overlapping k-mers

- Sequence
  - AACCGG
- K-mers (k=4)
  - AACC ACCG CCGG
- Graph

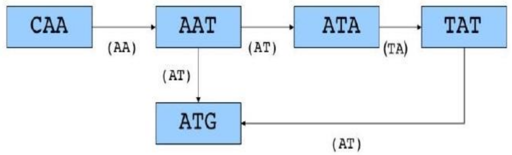
linear graph → one contig

- Sequence
  - AATAATA
- K-mers (k=4)
  - AATA ATAA TAAT <u>AATA</u> (repeat)
- Graph



cycle → contig breaks here

- Sequence
  - CAATATG
- K-mers (k=3)
  - CAA AAT ATA TAT ATG
- Graph



branching graph → two possible paths → ambiguity

Choosing k: the critical parameter

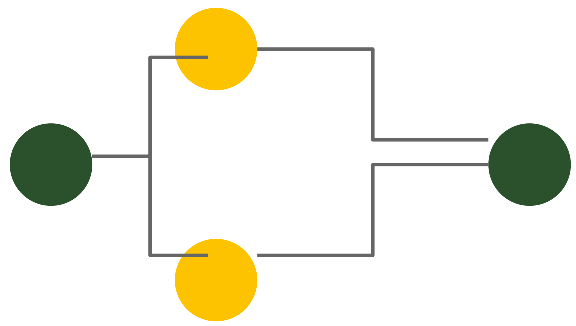
⚠️ **k too small**	Graph is over-connected. Many false edges, ambiguous paths. High memory. More errors incorporated.
⚠️ **k too large**	Graph becomes too sparse. Reads do not overlap sufficiently. Coverage gaps fragment assembly.
✔ **Optimal k (odd, < read length)**	multiple k values (21,33,55,77…)

LIMITATIONS

The central problem: repeats

A repeat longer than the read length cannot be traversed with confidence. In the assembly graph it appears as a branching loop

The assembler doesn't know how many times to go round, so it conservatively breaks the contig.



A repeat = a branch the assembler can't resolve

- **"Contigs end at repeats"** —rule of thumb
- **rRNA operons, IS, Tn (5–7 kbp, multi-copy)** fragment bacterial assemblies
- **E. coli (~4.6 Mbp)** → tens to hundreds of contigs with short reads

FROM READS TO A GENOME

A standard de novo assembly workflow

1. **QC reads**
   fastp, FastQC, Kraken2
2. **Assemble**
   SPAdes / Flye / Unicycler
3. **Assess**
   QUAST, BUSCO, CheckM
4. **Annotate**
   Prokka / Bakta

FASTQ → **FASTA (contigs)** → **GFF / GenBank**

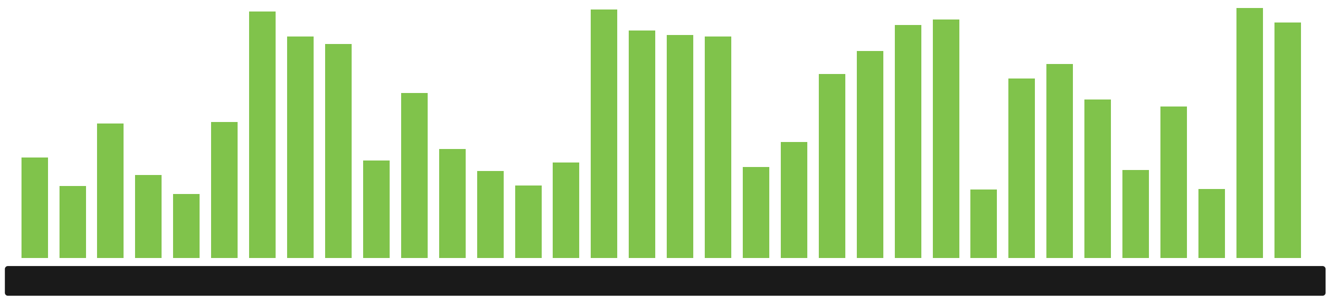
The assembler outputs a multi-FASTA — one entry per contig.

A 5 Mbp genome at 100× Illumina coverage assembles in minutes on a modest server.

Coverage

Coverage (depth) is the average number of reads spanning each base

Too little and the assembly fragments; far too much barely helps.



read depth across the genome

Rules of thumb

Illumina: 50–100×

(below 30× → more fragmentation)

Long reads: 30–60×

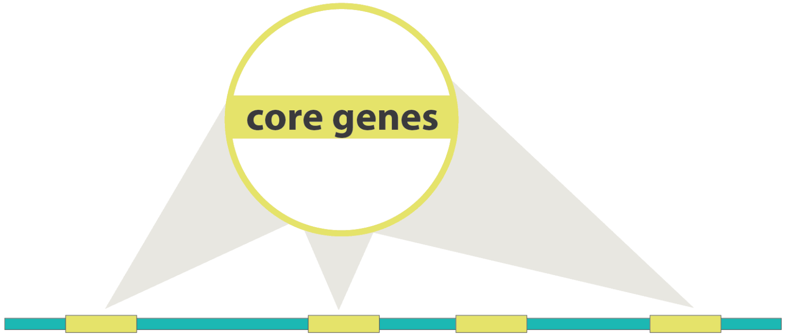
HiFi can close at lower depth

FROM READS TO A GENOME

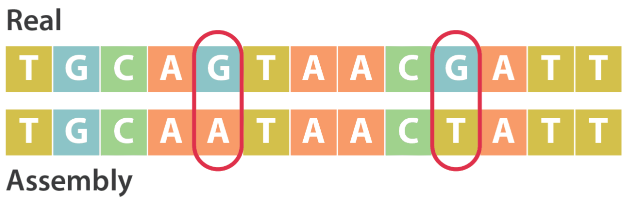
Assessing assembly quality (3C)

Contiguity

Completeness

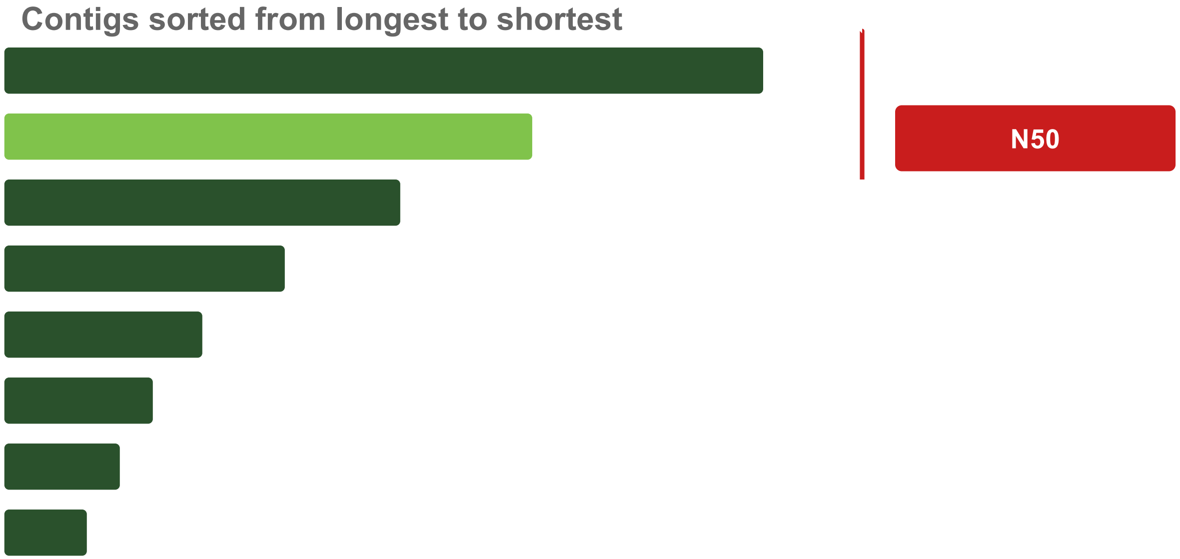


Correctness



Contiguity

Fewer, longer contigs are generally better. N50 captures this in one number.

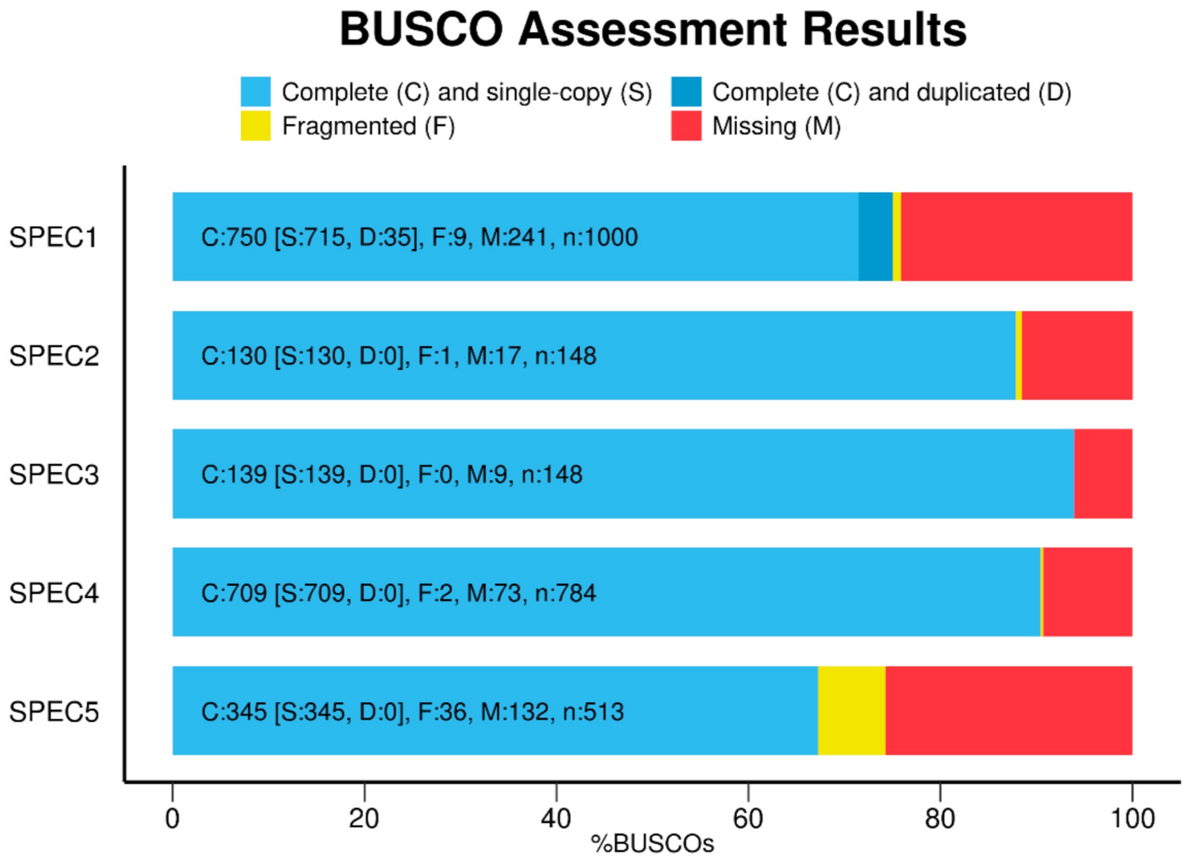


N50
the contig length at the 50% mark of cumulative length

Also report: number of contigs, total length (≈ expected genome size), largest contig.

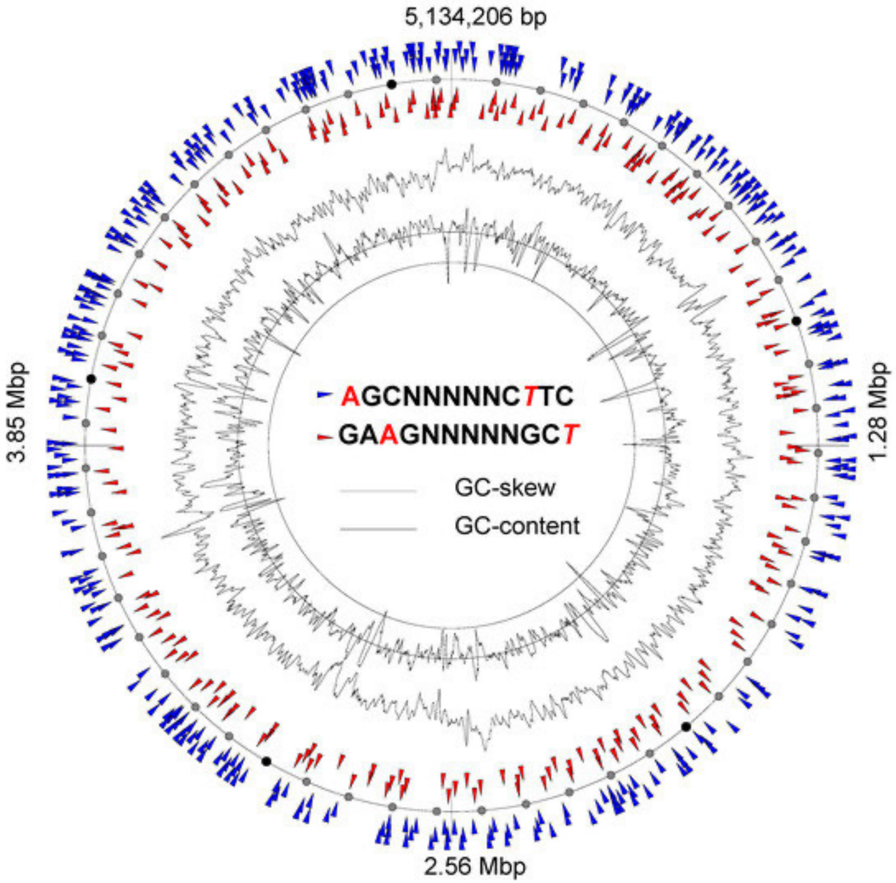
NG50 uses the expected genome size — comparable across species.

Completeness



BUSCO / CheckM — look for universal single-copy genes; ~100% = complete

Correctness

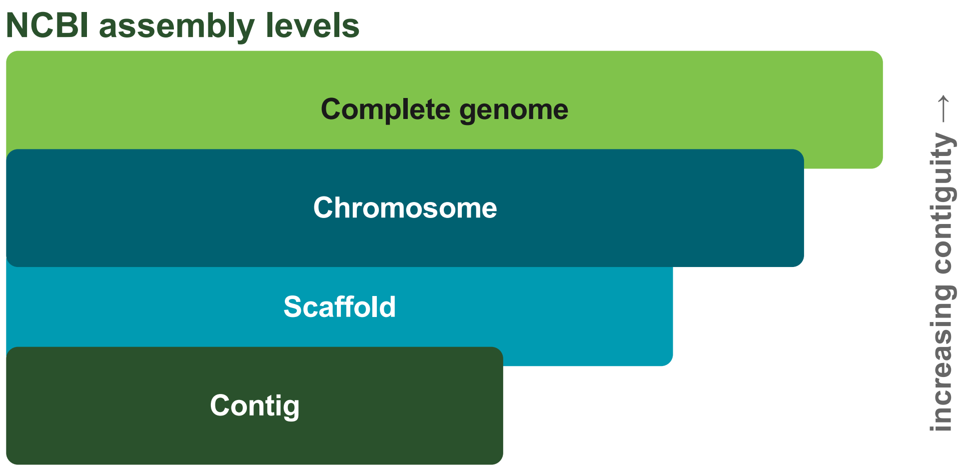


GC content — large deviation hints at contamination

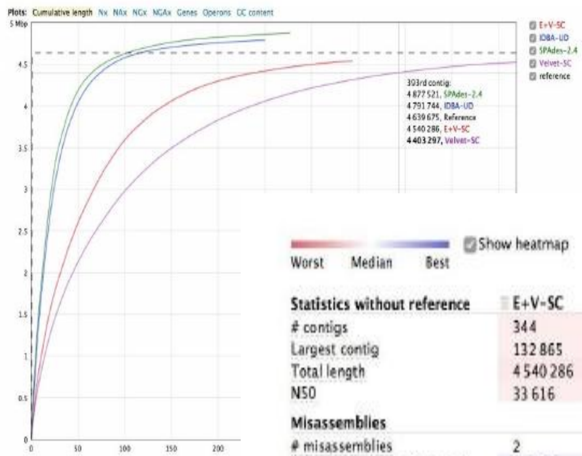
Expected number of rRNA operons or other known features — at expected locations

FROM READS TO A GENOME

Aseembly levels



QUAST generates a standardised quality report



Show heatmap

Worst Median Best

Statistics without reference	E+V-SC	IDBA-UD	SPAdes-2.4	Velvet-SC
# contigs	344	250	277	519
Largest contig	132 865	224 018	269 177	121 367
Total length	4 540 286	4 791 744	4 877 521	4 526 656
N50	33 616	96 947	106 927	20 445
Misassemblies				
# misassemblies	2	9	2	2
Misassembled contigs length	23 485	66 335	26 551	22 359
Mismatches				
# mismatches per 100 kbp	2.26	3.65	5.06	1.77
# indels per 100 kbp	0.7	0.2	0.7	0.92
# N's per 100 kbp	0	0	4.86	0
Genome statistics				
Genome fraction (%)	91.727	94.943	95.759	91.43
Duplication ratio	1.001	1.001	1.004	1.002
# genes	3767 + 160 part	4026 + 80 part	4046 + 102 part	3630 + 288 part
# operons	723 + 87 part	802 + 40 part	809 + 48 part	650 + 158 part
NGA50	32 051	96 947	110 539	19 791
Predicted genes				
# predicted genes (unique)	4258	4394	4417	4331
# predicted genes (>= 0 bp)	4258	4394	4490	4331
# predicted genes (>= 300 bp)	3643	3736	3784	3666
# predicted genes (>= 1500 bp)	524	559	559	515
# predicted genes (>= 3000 bp)	44	49	48	39

Extended report

Core metrics

N50

Contig length at the 50% cumulative assembly mark.

↑ *Higher = more contiguous*

Contigs

Total fragments. Ideal: 1 complete chromosome.

↓ *Fewer = more complete*

Largest contig

Size of the longest contig — can span repeats.

↑ *Higher for circular assemblies*

Genome fraction

% of reference covered. Requires a reference genome.

↑ *Should be ≥ 99%*

Misassemblies

Structural errors vs. reference (relocation, inversion, translocation).

↓ *Zero is ideal*

SCOPE

What de novo assembly does well

No reference needed

Indispensable for new, divergent or poorly studied organisms.

Full genome content

Captures accessory genes, plasmids and mobile elements.

New / Divergent pathogens

For which no reference is available.

MOTIVATION

Some applications

Gene-by-gene typing

MLST, cgMLST, wgMLST need the isolate’s allele sequences.

Serotyping in silico

Capsule and flagellar loci reconstructed from the assembly.

AMR & virulence in context

BLAST genes on contigs — see what element they sit on.

Plasmids & mobile elements

Reconstruct replicon boundaries and gene neighbourhoods.

Pangenome analysis

Compare gene content across hundreds of isolates.

SCOPE

Limitations — and “good” is task-dependent

Fragmentation —

short-read assemblies break at every long repeat.

Coverage & quality —

too little coverage or poor reads → broken, error-prone drafts.

Computational cost —

bacteria are quick; eukaryotes / metagenomes are heavy.

Mis-assemblies —

map reads back to check.

State the intended use before judging an assembly

SUMMARY

Key takeaways

- **Assembly is reference-free** — it reconstructs the isolate's own full genome
- **Long reads resolve repeats** — the route to closed, single-contig genomes
- **Coverage is king** — 50–100× Illumina; too little fragments the assembly
- **Judge with N50, BUSCO & QUAST** — but always against the intended use
- **Assembly and mapping are complementary** — typing & gene context vs high-resolution SNPs

Unicycler resolves chromosomes and plasmids by integrating graph-based assembly with long-read bridging

Pipeline architecture

1. **SPAdes:** De Bruijn graph from short reads
2. **Overlap graph:** Simplified contig connection graph
3. **Miniasm / Minimap2:** Long-read bridging across repeats
4. **Racon / Medaka:** Consensus polishing of bridged regions
5. **Circularisation:** Repeat resolution + chromosome closure

Three modes of Unicycler

Short-read only

```
unicycler -1 R1.fq -2 R2.fq -o out/
```

Illumina only · fast · high accuracy

⚠ ***Rarely circular — repeats unresolved***

Long-read only

```
unicycler -l long.fq -o out/
```

ONT or PacBio only · can close chromosomes

⚠ ***Low accuracy if error rate is high***

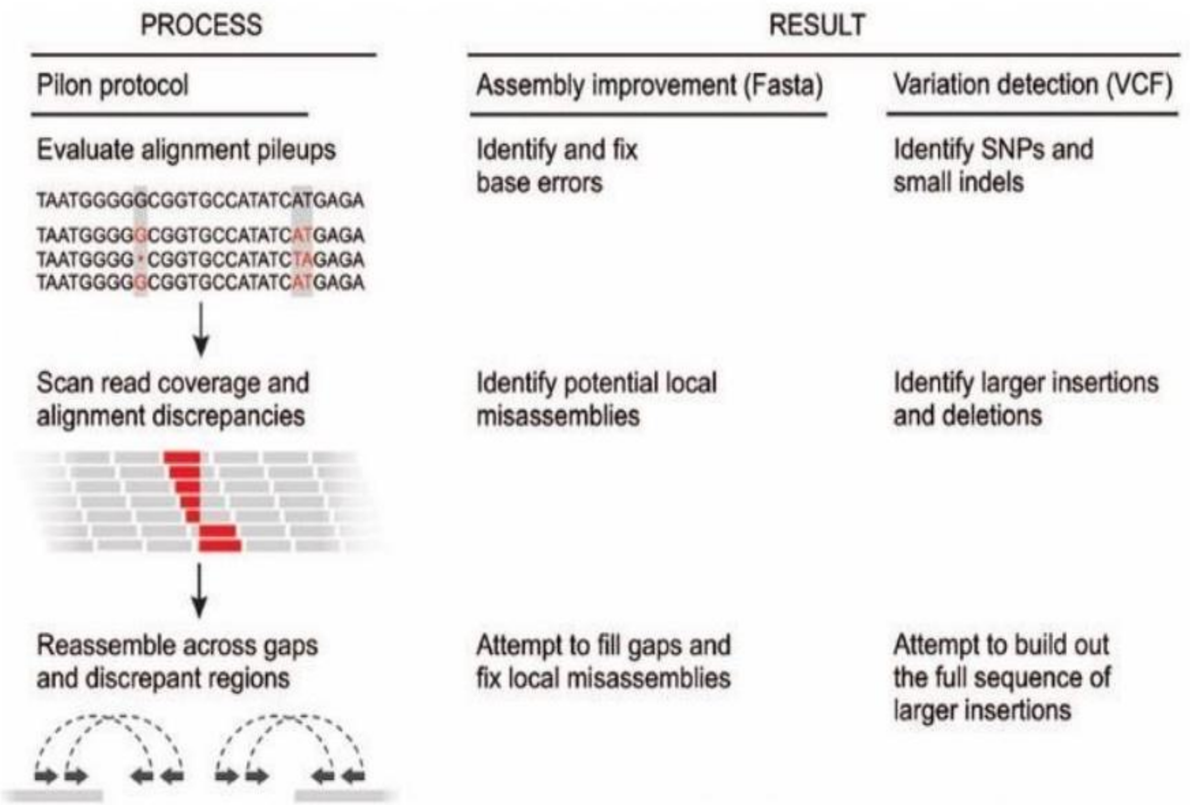
Hybrid (recommended)

```
unicycler -1 R1.fq -2 R2.fq -l long.fq -o out/
```

Illumina accuracy + ONT contiguity · complete closed genomes

✓ ***Gold standard for publication-grade genomes***

Polishing corrects systematic errors by realigning raw reads to the draft assembly and calling a new consensus



RACON — Rapid Consensus

Algorithm: Partial-order alignment (POA) graph. Builds a multiple-sequence alignment of overlapping reads, then extracts the highest-scoring path as consensus.

Input: Draft FASTA + original reads (FASTQ) + alignments (PAF/SAM via Minimap2).

Rounds: 2–4 iterations recommended. Each round re-aligns to the previous consensus; error rate drops ~10× per round.

Speed: Very fast (minutes). Fully parallelised with --threads.

Best for: First-pass polishing of ONT or PacBio CLR assemblies before Medaka.

Dragonflye streamlines long-read-only assembly

Dragonflye is a rapid long-read assembly pipeline (ONT / PacBio) wrapping Flye, Racon, Medaka, and Rasusa — a streamlined alternative to Unicycler hybrid when only long reads are available.

Internal tools

`Nanoq / NanoFilt`	Long-read quality filtering
`Rasusa`	Random coverage subsampling
`Flye / Raven`	Overlap-graph assembly
`Minimap2`	Read-to-assembly realignment
`Racon → Medaka`	Iterative consensus polishing

Workflow steps

01
Read QC
Nanoq

02
Assembly
Flye nano-hq

03
Polishing
Racon→Medaka

04
Evaluate
Bandage+QUAST

Dragonflye vs Unicycler Hybrid

Criterion	Dragonflye	Unicycler Hybrid
Speed	★★★★★	★★★
Quality (long only)	★★★	★★★★
Requires Illumina	No	Yes
Ease of use	High	Medium
Best for	Field ONT	Final publication

ASSEMBLY COMPARISON

Same *S. aureus* JKD6159 isolate and reference; student tasks differ by assembler workflow.

EXPERIMENTAL DESIGN

ILLUMINA — s100 / s200 / s300 × 500k paired reads

ONT — s100 / s200 / s300 × 2,500 / 5,000 / 10,000 reads

Unicycler

STUDENT TASK: assemble, then run QUAST

Short only	Illumina → SPAdes graph assembly → assembly.fasta
Long only	ONT → miniasm (fast, lower accuracy) → assembly.fasta
Hybrid	Illumina + ONT → SPAdes graph + bridging → assembly.fasta

Dragonflye

PRE-PREPARED OUTPUTS: run QUAST only

Long only	Provided ONT → Flye assembly → contigs.fa
Hybrid	Provided hybrid → Flye + 1× Pilon + 1× Polypolish → contigs.fa

Use supplied Dragonflye contigs: run QUAST only

QUAST: evaluate the assigned contig files

N50	**Contigs**	**Misassemblies**	**Reference covered**
contiguity	fragmentation	structural accuracy	completeness

thanks!

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